another physical dimension. To Them the past might be a canyon They can climb into and the future a mountain They can climb up."

Einstein's laws, extended into the bulk, tell us that local bulk time can't behave this way. Nothing in the bulk can go backward in local bulk time. However, when looking into our brane from the bulk, Cooper and bulk beings can and do see our brane's time (bedroom time) behave like Brand says. *As seen from the bulk*, "our brane's time can look like just another physical dimension," to paraphrase Brand. "Our brane's past looks like a canyon that Cooper can climb into [by traveling down the tesseract's diagonal channel], and our brane's future looks like a mountain that Cooper can climb up [by traveling up the tesseract's diagonal channel; Figure 29.14]."

This is my physicist's interpretation of Brand's words. And Chris interprets them similarly.

Touching Brand Across the Fifth Dimension



In *Interstellar*, with the quantum data safely in Murph's hands, Cooper's mission is finished. The tesseract, carrying him through the bulk, begins to close.

As it is closing, he sees the wormhole. And within the wormhole, he sees the *Endurance* on its maiden voyage to Gargantua. As he sweeps past the *Endurance*, he reaches out and gravitationally touches Brand across the fifth dimension. She thinks she has been touched by a bulk being. She has . . . by a being riding through the bulk in a rapidly closing tesseract. By an exhausted, older Cooper.